Comparative analysis of EKF and Particle Filter performance for an acoustic tracking system for AUVs exploiting bearing-only measurements

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***Abstract*—This paper aims to compare different navigation state estimation approaches for the tracking of an Autonomous Underwater Vehicle (AUV) based on Direction of Arrival (DoA) measurements. The use of passive acoustic devices to compute the DoA information allows for a high rate (up to 1Hz in the proposed tests) of the measurements while the AUV moves underwater. The informative content of these data integrated with the synchronised data of the on-board sensors allows for a fine tracking of the vehicle under test for navigation performance evaluation. The paper addresses the issue of the selection of the proper data filtering strategy for the estimation of the navigation state of the AUV. In particular, it investigates whether the adoption of a Particle Filter (PF) has significant advantages with respect to a classical approach based on an Extended Kalman Filter (EKF). The two approaches exploit the same models and parameters to limit the comparison to the filtering strategy. The ground truth is the GPS signal provided by the receiver integrated within the antenna on the AUV under test. The comparative analysis considers both the estimation performance and the computation load. Results are extensively provided for a subset of all the processed dataset, all the remaining part of data confirm the reported analysis.**

***Index Terms*—Kalman filter, Particle filter, Autonomous Underwater Vehicles, Acoustic tracking system, Bearing-only measurements, Acoustic Modems, USBL**

I. Introduction

Autonomous Underwater Vehicles (AUVs) are nowadays widely used in marine applications, thanks to the technological progress over the years that has made such vehicles easier to use and less expensive than in the past. The absence of GPS in sub-sea scenarios affects the vehicle underwater navigation performance which negatively impacts the goodness of the sub-sea gathered data. For this reason, the underwater vehicle localisation is a very challenging topic. In literature, a lot of research works deal with this issue. A review of the state of the art of AUV navigation and localisation can be found in [1], where the Extended Kalman Filter (EKF) and Particle Filter (PF) approaches are presented in Simultaneous Localisation and Mapping (SLAM) applications. Different PF applications can be found in [2] and [3]. The work presented in [2] shows a PF approach based on time difference of arrival (TDOA) of acoustic signals from beacons around a robot. Reference [3] exploits environment semantic information to extend the standard particle filter approach. In previous works of the authors, target tracking algorithms based on bearing only measurements provided by Ultra-Short Baselines (USBLs) sensors and calibration procedure of an acoustic tracking system have been implemented, in order to verify the vehicle conformity with long term underwater navigation. In particular, the authors aimed at creating an Underwater Test Range (UTR) system for underwater vehicles. This system is in charge of tracking the vehicle position using filters which expoloit Direction of Arrival (DoA) measures provided by USBLs. The tracking system is detailed in [4], and the utilised calibration procedure utilised is presented in [5]. In a further past work, the authors have used the Unscented Kalman Filter (UKF) based approach for resolving the localisation issue in the UTR system [6]. The purpose of this work is to compare EKF and PF state estimation approaches for AUV tracking

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using DoA measurements. The choice of these filters is due to the non-linearity of the problem. In addition, the chosen filters are widely used for localisation problems both in maritime and non-maritime scenarios. In this paper, the measurements from two USBL devices have been utilised. These devices have been used only as passive sensors, in order to detect the vehicle transmitted signals. In this way the USBLs provide only DOA measures. The advantage of this configuration is the high vehicle ping rate which guarantees frequent DoA observations. The experimental campaign planned in order to acquire the exploited data, was performed on March 2019 in La Spezia (Italy) using an eFolaga AUV.

The paper is structured as follows: section II presents the navigation model, section III describes the experimental setup used during the sea test, section IV reports the comparison results in terms of computational time, position, velocity and drift estimation goodness and finally section V summarises conclusions.

II. Models

In this work, the localisation problem is limited to the horizontal plane thanks to the reliable depth measurements provided by the sensor installed on the vehicle. The state used for the estimation filters is composed of:

- p_{north}, the position of the vehicle along the North direction in a NED reference frame;
- p_{east}, the position of the vehicle along the East direction in a NED reference frame;
- v, the speed of the vehicle along the course direction;
- δ, the drift of the vehicle, which is the angle between heading and course directions;

Thus, the state vector can be expressed as:

$$x = [p_{north}, p_{east}, v, \delta]^T \in \mathcal{R}^4$$

The model has as inputs the heading measurements provided by the magnetic compass, which is used in the filters prediction step. The state-transition model represents a model with slowly variable speed along the course axis and is defined by the following equations:

$$\begin{cases} p_{north_{k+1}} &= p_{north_k} + v\cos(\psi+\delta)\, dT \\ p_{east_{k+1}} &= p_{east_k} + v\sin(\psi+\delta)\, dT \\ v_{k+1} &= w_{v_k}\, dT \\ \delta_{k+1} &= w_{\delta_k}\, dT \end{cases} \tag{1}$$

where ψ is the heading angle used as input, w_v and w_δ are the white Gaussian process noise which characterise the derivative with respect to the time of the state respective component. The time interval dT has a value of $0.1s$, as the filter loop rate is $10Hz$, which corresponds to the maximum vehicle sensor acquisition rate.

The observation model, which exploits the DoA measurements provided by two USBLs, is:

$$h = atan_2(p_{east} - s_{east}, p_{north} - s_{north}) \tag{2}$$

where $atan_2$ returns the four-quadrant inverse tangent, s_{north} and s_{east} are the positions of the USBL providing measures in NED reference frame, obtained through the calibration procedure presented in [5] . In the EKF the observation model is used in the correction step; in the PF the DoA measurements are exploited to compute the likelihood of each particle within each correction step and to consequently update their associated weights.

III. Experimental Setup

A. Operational scenario

This investigation was performed in post-processing, exploiting part of the dataset acquired during a campaign performed in La Spezia (Italy) during March 2019 at the SEALab, the joint research laboratory between the Inter-university Center of Integrated Systems for the Marine Environment (ISME), of which University of Pisa (UNIPI) is a co-founder, and the Naval Support and Experimentation Centre of Italian Navy (CSSN).

The working area is an artificial basin between two piers, within which two USBLs have been deployed as DoA sensors. It is a challenging environment from an acoustic point of view because of the very shallow water and the proximity to the quayside which frequently cause the multi-path phenomena generating unreliable measurements.

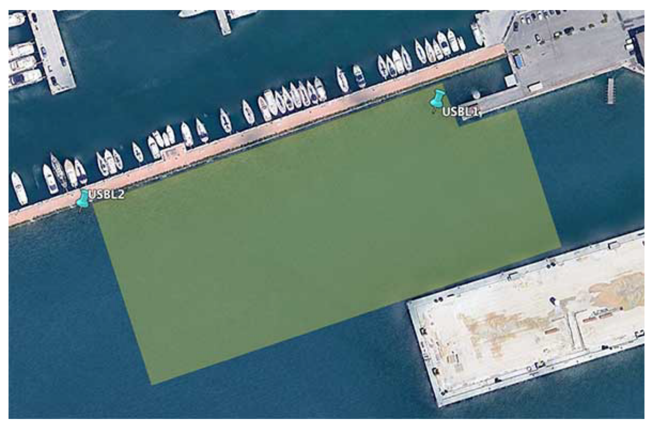


Fig. 1: Working area in green with the position of the two USBL devices

Figure 1 shows the operational scenario and the USBL devices positions. The green polygon represented in the figure defines the working area where the vehicle was moving during the tests.

B. Equipment setup

The trials were carried out using the eFolaga AUV [1], showed in Figure 2, a torpedo-shape vehicle made up of a pair of fibreglass cylinders connected to two terminating wet sections where propelling thrusters and pump jets are located, together with a bladder enabling variations of the vehicle buoyancy. For these trials the vehicle was equipped with: depth sensor, compass, 2D inclinometer and a GPS. All the sensors data are collected at a rate of 10 Hz, except for the GPS

Fig. 2: eFolaga AUV

which provides measures at 1 Hz. The vehicle was able to communicate with the USBL devices thanks to an acoustic modem (Evologics S2C M 18/34) integrated on-board. The acoustic tracking system, was composed of two DoA sensors, in order to track the vehicle. During the tests, the DoA sensors involved were USBL devices deployed in known position on the seafloor properly calibrated in post-processing to determinate their attitude [5]. The vehicle, equipped with a compliant modem, was programmed to transmit in broadcast acoustic packets containing current timestamp with a frequency of 1 Hz. The USBL devices receiving the message are likely able to compute the DoA, and to associate the vehicle timestamp to the measurements by reading the content of the message. Thus, this information can be easily synchronised in post-processing with other data collected on-board the vehicle thanks to the timestamp encoded in the acoustic messages.

C. Mission description

In this work the dataset is used for the comparison analysis. Such dataset was acquired during the test performed in the scenario detailed in III-A with the equipment presented in III-B, is used for the comparison analysis. This dataset is composed of six different missions that include similar manoeuvres, in order to carry out a consistent analysis.

Figure 3 shows an example of trajectory, displaying the GPS measurements, performed by the vehicle in the missions analysed. The position estimation are performed exploiting missions containing vehicle trajectories on the surface, in order to utilise as ground truth the available latitude and longitude measurements provided by the GPS device mounted on the vehicle.

IV. Results

The authors aim to evaluate the performances of EKF and PF filters in terms of computational time and estimation performance. The estimation evaluation is based on the subset of missions that took place on the surface. The GPS measurements available for these missions have not considered within the filters, except for the initialisation, but have been used as Ground Truth to quantify the estimation performance. The comparative analysis is carried out estimating the state vector by varying: particles number in the PF, missions and

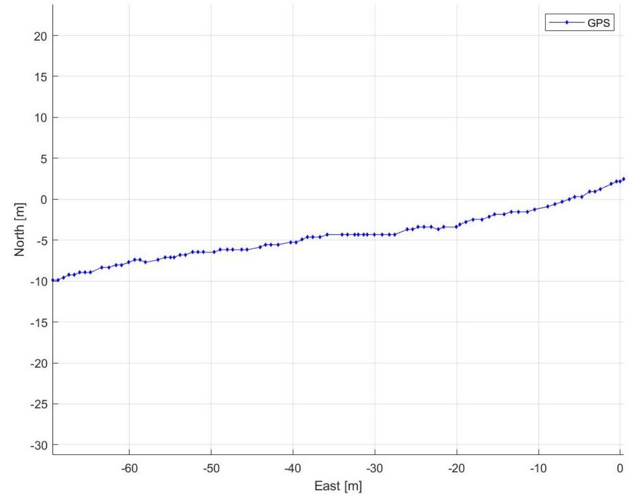


Fig. 3: Example of mission

process noise on the drift state in both EKF and PF filters. In particular:

- **mission**: in order to get a clearer picture of the performance of the two filters, six different missions, presented in the subsection III-C, are used. All of these are extracted from the investigation detailed in the subsection III-A performed with the equipment showed in the subsection III-B. Each mission represent a different vehicle trajectory, performed in the same conditions.
- **process noise on the drift**: the values (expressed in deg/s) imposed on the drift standard deviation are,

$$\delta_{std\_dev} = [3, 5, 10, 15] \tag{3}$$

- **number of particles**: in the PF the number of particles vary as follows,

$$\#_{particles} = [500, 1000, 3000, 5000] \tag{4}$$

The state estimated by each filter, and the corresponding computational time, for each configuration of parameters, were stored and post-analysed in order to extract information in terms of quality, robustness and efficiency. In this section the results relating to the performed analysis will be presented. For space issues, only two missions will be shown, as the results obtained are similar for all six missions of the analysed dataset

A. Position estimation error analysis

The position estimation error is evaluated through the Root Mean Square Error (RMSE) respect to GPS measurements in meters and evaluating the estimation error over the time, also in meters, for each mission and each possible combination of the parameters defined above.

Figure 4, 5, 6 and 7 show the estimation error over the time of EKF (in black in the figure) and PF (in red in the figure) with respect to GPS measurements used as reference, in two different conditions of the process noise on the drift state. These plots show a PF performance improvement when

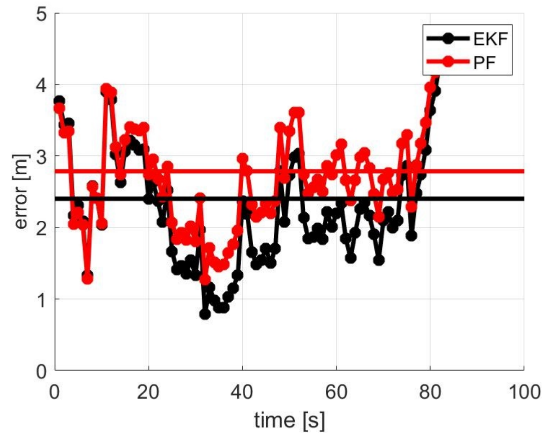


Fig. 4: Mission 1. Error over the time of EKF and PF estimation compared with the GPS ground truth. The process noise on the drift is 3 deg/s. The number of particles is 1000.

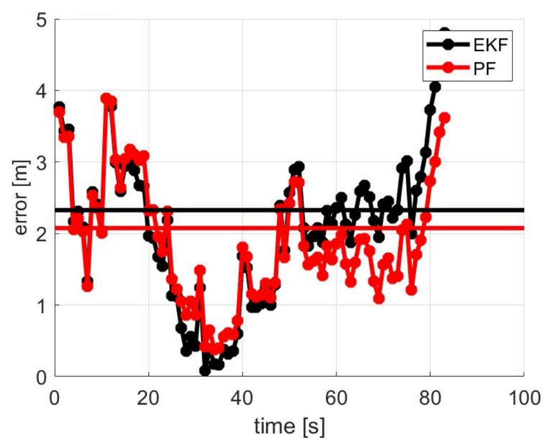


Fig. 5: Mission 1. Error over the time of EKF and PF estimation compared with the GPS ground truth. The process noise on the drift is 15 deg/s. The number of particles is 1000.

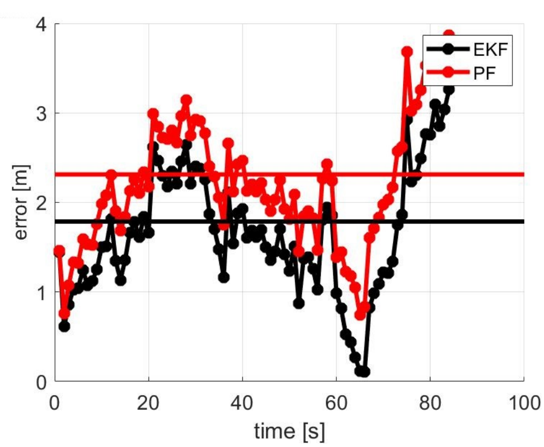


Fig. 6: Mission 2. Error over the time of EKF and PF estimation compared with the GPS ground truth. The process noise on the drift is 3 deg/s. The number of particles is 1000.

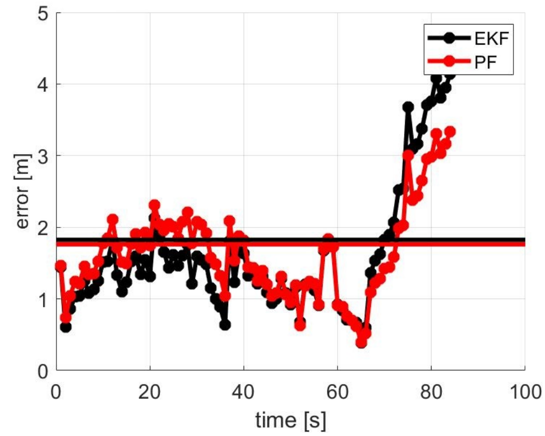
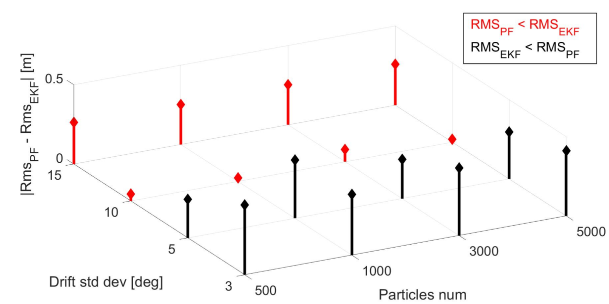


Fig. 7: Mission 2. Error over the time of EKF and PF estimation compared with the GPS ground truth. The process noise on the drift is 15 deg/s. The number of particles is 1000.

the process noise on the drift increases. In fact, the average RMSE of the PF estimation is lower when the drift standard deviation is 15 $\frac{deg}{s}$ (figures 5 and 7). In the reported cases the standard deviation of the drift noise are 3 and 15 $\frac{deg}{s}$ respectively. In this paper, the data of only two mission are reported, but this trend has been verified in all the six missions in the analysed dataset.



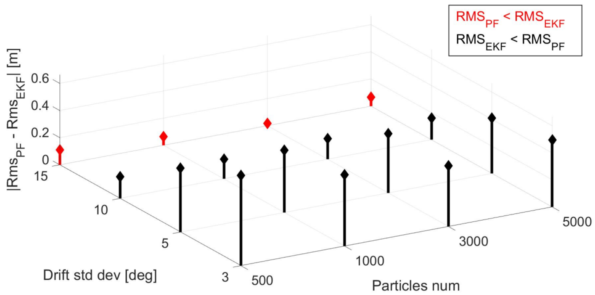
(a) *Stems of absolute value of the difference between RMSE for each configuration*

Drift std dev [°/s]					
	15	2.06	2.07	2.07	2.06
	10	2.29	2.31	2.26	2.32
	5	2.36	2.36	2.36	2.36
	3	2.42	2.42	2.42	2.42
		500	1000	3000	5000
		Particle numbers			

(b) *Table representing the lower RMSE for each configuration*

Fig. 8: Mission 1. RMSE for each configuration correlated by an illustrative table.

Figure 8 and 9 report the absolute value of the difference between the EKF and PF RMSE on the vehicle position and a table which quantifies the lower RMSE value for each configuration, for mission 1 and 2 respectively. The cases in which the EKF estimates are better than the ones obtained through the PF are displayed in black, the others in red. Also



(a) *Stems of absolute value of the difference between RMSE for each configuration*

Drift std dev [°/s]	500	1000	3000	5000
15	1.71	1.76	1.80	1.76
10	1.82	1.82	1.82	1.82
5	1.79	1.79	1.79	1.79
3	1.78	1.78	1.78	1.78
	Particle numbers			

(b) *Table representing the lower RMSE for each configuration*

Fig. 9: Mission 2. RMSE for each configuration correlated by an illustrative table.

this kind of data analysis based on the RMSE reveals the same trend: the PF performance increases when the process noise on the drift augments.

B. Drift and velocity analysis

In this subsection the estimation of the drift and velocity state by varying the process noise on the drift will be reported. Since the evolution of drift and velocity are similar by varying the missions, the results of only one is shown. Figures 10 and 11 show the evolution of the drift estimation, for EKF and PF respectively, by varying the process noise on the drift. Under the same process noise conditions, the drift estimated by the PF has greater fluctuations with respect to the one estimated by the EKF. Figures 12 and 13 instead, show that this aspect is not found in the velocity estimation evolution. Indeed, the two estimated velocity plots (EKF in figure 12) and (PF in figure 13) have the same trend.

C. Computational time analysis

This subsection will report an efficiency analysis in term of computational time between for EKF and PF filters. This analysis has been performed using a machine with an Intel Core i7-8550U CPU @1.99GHz in MATLAB environment.

The time analysis, reported in figure 14, represents the time required, for both filters, to estimates the vehicle positions of the entire considered mission. The filter estimations reported are composed of 943 prediction steps and 89 correction steps. The computation time measured, and showed in figure 14, are:

- **EKF**: 0.169 seconds
- **PF with 500 particles**: 0.743 seconds
- **PF with 1000 particles**: 1.071 seconds
- **PF with 3000 particles**: 1.517 seconds
- **PF with 5000 particles**: 1.989 seconds

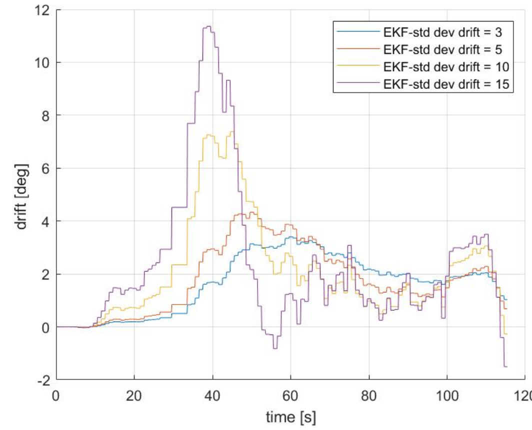


Fig. 10: EKF estimation drift plot over the time by varying the noise on the drift.

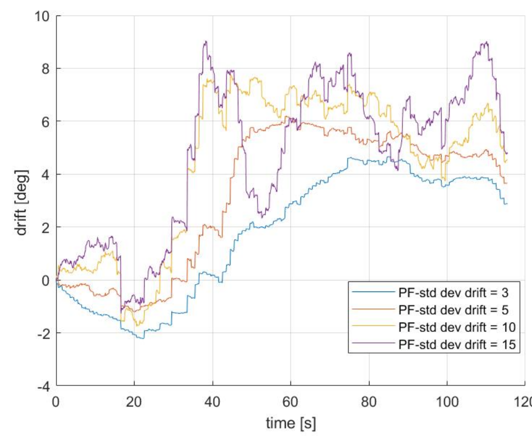


Fig. 11: PF estimation drift plot over the time by varying the noise on the drift. The particle number is set to 1000.

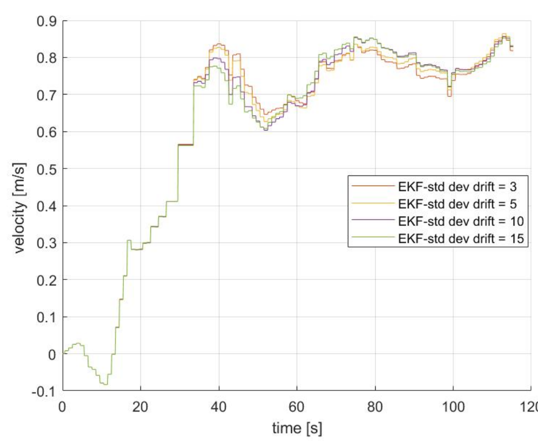


Fig. 12: EKF estimation velocity plot over the time by varying the noise on the drift.

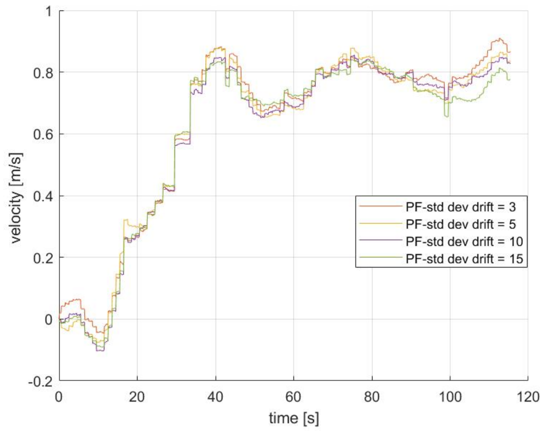


Fig. 13: PF estimation velocity plot over the time by varying the noise on the drift. The particle number is set to 1000.

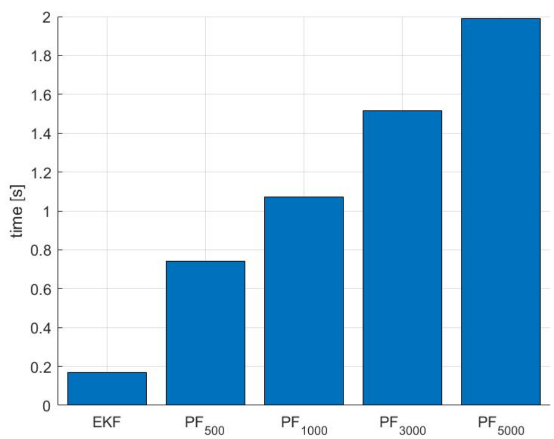


Fig. 14: Computational time analysis

As expected, EKF has higher performance than PF from a computational time point of view, and the process time of the PF augments when increasing the particles number. This result is also obtained from the analysis of all the other missions in the considered dataset; detailed results of only one mission are reported because of space issues.

V. Conclusions

A performance comparison between the Extended Kalman Filter and the Particle Filter has been reported in this paper for the particular scenario of AUV tracking in terms of trajectory estimation errors and computational time. This analysis revealed, as expected, a significant difference in favour of the EKF from the computational time point of view. Such disparity has been quantified for different numbers of particles in the PF. From a trajectory estimation point of view, the reported analysis, based on the RMSE, highlights a trend according to which the PF tends to improve as the standard deviation of the process noise on the drift state component increases. The difference on the RMSE always in the order of 10^{-2} meters does not suggest particular advantages in the use of an approach with respect to the other one for the specific application and scenario.

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